

9(a)

any **seven** from:

- 1 reaction / photosynthesis, controlled / affected, by more than one / several, factors ;
- 2 factor / variable, nearest its minimum value / in shortest supply ;
- 3 which, prevents increase in rate / restricts rate / makes rate plateau ;

to increase crop yield in glass house:

- 4 light intensity + temperature + carbon dioxide concentration ;
- 5 artificial lights / lamps / provide lighting at night ;
- 6 heaters / open windows / control ventilation ;
- 7 CO₂ from, cylinders / dry ice / combustion / (named) fuel heater ;
- 8 automation / computerised control / digital control ;
- 9 *ref. to optimum*, light intensity / temperature / CO₂ concentration / conditions ;
- 10 AVP ; e.g. water / irrigation / sprinklers
control / optimum, humidity
(named) fertiliser / nitrate / manure

7

Question	Answer	Marks
9(b)	any eight from: <i>meiosis I</i> 1 chromosomes, condense / thicken / spiralise ; 2 homologous chromosomes pair up / bivalents form ; 3 crossing over / chiasma(ta) ; 4 spindle fibres / microtubules, attach to / pull, centromeres / kinetochores ; 5 bivalents, line up / arranged, on equator ; 6 independent / random, assortment (of homologous pairs) ; 7 chromosomes, move / pulled, to poles ; <i>meiosis II</i> 8 (single) chromosomes / pairs of chromatids, line up on, equator / mid-line ; 9 at right angles to first equator ; 10 centromeres divide ; 11 chromatids separate ; 12 <i>ref. to</i> haploid / chromosome no. halved / one set of chromosomes ;	8

Question	Answer	Marks
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